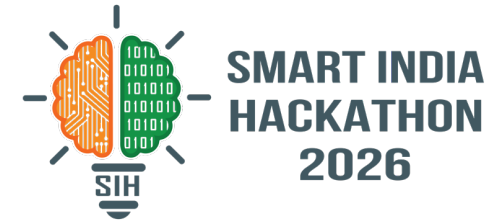


SMART INDIA HACKATHON 2026



TITLE PAGE

Problem Statement ID – SIH26231

Problem Statement Title – Digital Companion for Field Drug Testing

Theme – MedTech / BioTech / HealthTech

PS Category – Software

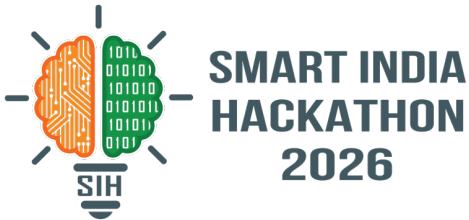
Team ID – «TEAM ID»

Team Name – «TEAM NAME as registered on the portal»





THE RECORD IS THE PRODUCT



not the colour reading — that part is an undergraduate exercise

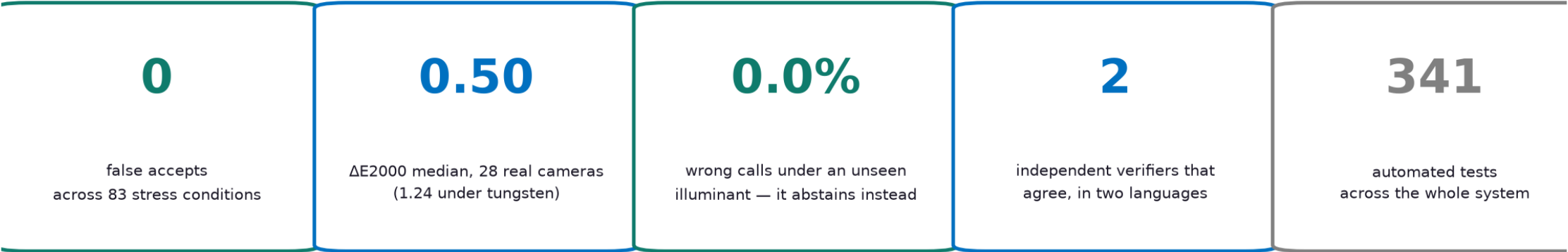
THE PROBLEM IS NOT THE COLOUR. The problem statement says it plainly: a field test "leaves no verifiable record... outcomes cannot presently be relied upon as documentary evidence." Reading a colour is an undergraduate exercise — and the naive app already ships commercially (DetectaChem MobileDetect).

OUR SOLUTION. The measurement is bound, at the instant of capture, into a hardware-attested, append-only record that anyone can re-derive offline — and that emits a Bharatiya Sakshya Adhiniyam Section 63 certificate and hands off to eSakshya / CCTNS-2.0.

WHAT IS ACTUALLY NEW:

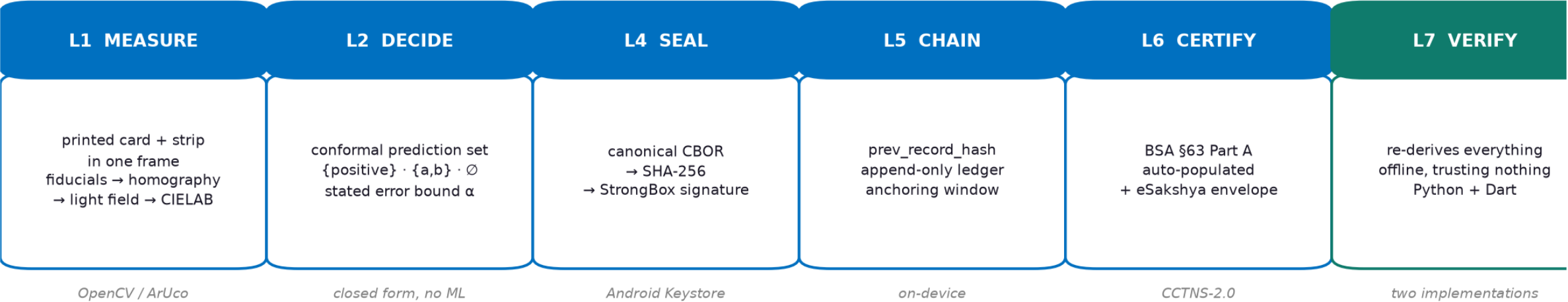
- Kit-agnostic — reagent is declared, not read from a vendor's serialised pouch, as the PS requires
- "Inconclusive" carries a stated error bound (conformal prediction), not a softmax confidence a court cannot cross-examine
- Two independently written verifiers that agree — a single implementation agreeing with itself proves nothing
- The statutory certificate is emitted, not left as an exercise

No new hardware: a printed colour card and the phone already issued.



TECHNICAL APPROACH

Flutter (Android) · OpenCV/ArUco · CIELAB + CIEDE2000 · conformal prediction · canonical CBOR + SHA-256 · ECDSA P-256 in StrongBox/TEE · verifiers in Python and Dart



No new hardware. A printed colour card and the phone already issued.

Measured on 28 real camera sensitivities × CIE illuminants: 0.50 ΔE_{2000} under D65, 1.24 under tungsten · 83-condition sweep: 0 false accepts

WHY DETERMINISTIC, NOT A NEURAL NETWORK: a least-squares colour transform is auditable and defensible in court. We ablated a learned classifier against it on 8,064 measurements rendered from 28 measured camera sensitivities — the closed-form score won on every split, so there is no learned component and no TFLite dependency.

FEASIBILITY AND VIABILITY

FEASIBLE TODAY: no new hardware, works with kits already purchased, runs fully offline, and a working core already exists — 341 automated tests, both verifiers agreeing on every record.

RISK	MITIGATION / HONEST STATUS
Replay: photographing a quality print or screen	NOT SOLVED — declared a residual risk. Needs multi-frame parallax.
No reagent dataset exists publicly	Surrogate colour ladders. Substituting NCB standards is a data change, not architecture.
StrongBox absent on some issued handsets	TEE fallback; the weaker guarantee is written into the record, not hidden.
Backdating inside the unanchored window	Hash chain + anchoring. The open window is stated on screen and in the record.
Bad light producing a confident wrong answer	Quality gate + conformal abstention. 0 false accepts across 83 conditions.

We list the attack we have NOT solved. A threat model that overstates its defences is worth less than none — a judge who finds the gap first concludes we never looked.

IMPACT AND BENEFITS

Turns a presumptive field test from an unusable note into an artefact a court can test — without building a second evidence store for the State to maintain.

NCB / POLICE	COURTS & THE ACCUSED	COST & REACH
• A field test becomes documentary evidence • §63 Part A auto-filled at capture • Plugs into eSakshya / CCTNS-2.0 — no new silo • Works with kits already purchased • Offline queue — no coverage required	• Anyone can verify offline, trusting no one • Disagreements recorded, not suppressed • 'Inconclusive' has a stated error bound • Verifier says what it CANNOT prove • Same tool serves defence and prosecution • Records cannot be deleted or reordered	• One printed card $\approx$ ₹5, matte A4 • No new hardware, no new procurement • Runs fully offline — NDPS field reality • Any Android handset already issued • Kit-agnostic: not tied to one vendor • Card reprints locally, no supply chain

Deliberately NOT built: a confirmatory test, a national database of results, or any scoring of persons. CCTNS-2.0 stays the system of record; a parallel silo would add a surveillance surface and a liability for no benefit.

STATUTE & GOVERNMENT SYSTEMS

Bharatiya Sakshya Adhiniyam 2023, Section 63 + the Schedule (certificate, Parts A and B; hash value and algorithm required) · indiankanoon.org/doc/125020475

NDPS (Seizure, Storage, Sampling and Disposal) Rules 2022 — GSR 899(E) · eSakshya, NCRB / CCTNS-2.0 · informatics.nic.in

PRIOR ART — CONCEDED, NOT IGNORED

DetectaChem MobileDetect: smartphone + colorimetric pouches, GPS-tagged PDF reports. The naive solution is a shipping product; our claim is narrower and sits in the binding, the abstention bound and the verifiability.

METHOD

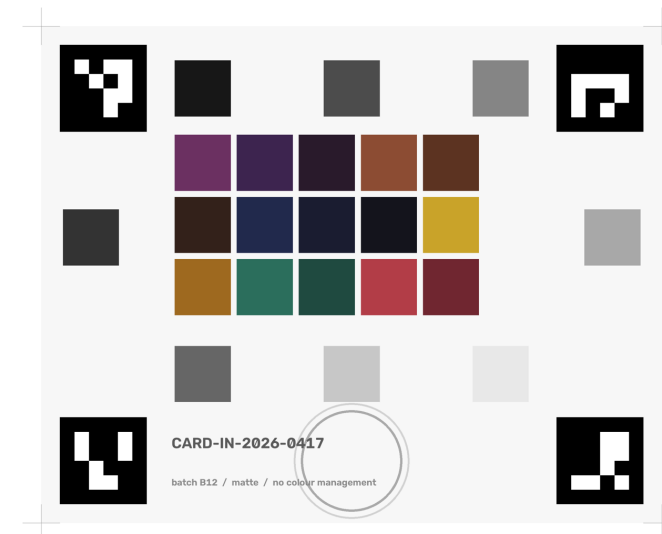
Finlayson et al., root-polynomial colour correction (exposure-invariant) · Sharma, Wu & Dalal (2005), CIEDE2000 conformance data · Vovk et al., conformal prediction · Android key attestation & StrongBox (AOSP)

MEASURED DATA USED IN OUR EVALUATION

Jiang, Liu, Gu & Süssstrunk (2013) camera spectral sensitivity database, 28 cameras — Zenodo 3245883, CC BY-NC-SA 4.0 · CIE illuminant SPDs, 1931 observer and ColorChecker reflectances via colour-science (BSD-3)

STATED LIMITATION

No public dataset of colorimetric drug-test images exists, and no spectral library of NDPS reagent developments. Our classifier is calibrated on surrogate colour targets. Substituting NCB reagent standards is a DATA change, not an architecture change: L1 and L3–L7 are substance-independent by construction.



The only physical artefact: a matte A4 print — fiducials, 15 reagent-gamut patches, 8 neutrals, reaction well.

REPRODUCIBILITY

Working code, 341 automated tests, and every figure on these slides regenerable from one command.

Every number here is measured on synthetic frames or measured spectral data — none is a field accuracy. The physical capture set is the critical path, and we say so.